PROSPECTS FOR THE DEMILITARIZATION OF OUTER SPACE: FROM "SOFT REGULATION" TO "HARD" TREATY MECHANISMS?

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ABSTRACT

The 1967 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies[1] (Outer Space Treaty), establishes the obligation of its parties to carry out space activities in the interests of maintaining international peace and security and promoting international cooperation and understanding. It does not, however, prohibit military activities in outer space, if it is not contrary to the Charter of the United Nations (UN). The Outer Space Treaty establishes a regime for the partial demilitarization of outer space, prohibiting: any objects with nuclear weapons or any other weapons of mass destruction from being placed in orbit around the Earth; the placement of such weapons on celestial bodies; and the stationing of such weapons in outer space in any other way. At the same time, the Moon and other celestial bodies are fully demilitarized.

Since the adoption of the Outer Space Treaty, there have been many attempts to limit the military use of space by concluding additional international agreements. This was done even during the Cold War era when Great Power competition between the United States and the Union of Soviet Socialist Republics was at its height. In the modern era, efforts are being made to ensure the demilitarization of outer space via the drafting and conclusion of a special international treaty. Unfortunately, these efforts at international

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[1] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty].

treaty-making have, thus far, been unsuccessful. This has led proponents of demilitarization to seek to achieve their goals in other ways: by putting forward new or alternative initiatives, developing and adopting acts of "soft law" and encouraging the passage of resolutions in the United Nations General Assembly, regional international organizations and the like. The expert community offers the doctrinal model parameters for the implementation of military activities in outer space.

In the context of the multi-vector development of regulators restricting military activities in space the question arises about the prospects for strengthening the regime of its demilitarization in the near future.

I. INTRODUCTION

I have enthusiastically accepted the offer of Sa'id Mosteshar, Director of The London Institute of Space Policy and Law, to contribute to this issue of the *Journal of Space Law* dedicated to commemorating Professor Bin Cheng's great contribution to the development and understanding of international space law. My teacher and friend Professor Gennady P. Zhukov,[2] the patriarch of Russian doctrines of international space law, succinctly described to me the seminal nature of Professor Cheng's *Studies in International Space Law* (1997), as one of Cheng's most remarkable works.[3] Zhukov emphasized the importance of this book's comprehensive analysis of the development of space law, from the pre-satellite period to the present day. According to Zhukov, special attention should be paid to Cheng's interpretation of the concept of space exploration for peaceful purposes, which is consistent with the understanding of

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[2] Gennady Petrovich Zhukov (1924-2014) is a doctor of law, professor, honored lawyer in Russia. In 1966, he defended the first thesis of the Doctor of Law in the USSR on international legal problems of space exploration. Zhukov was an academician of the International Academy of Astronautics since 1965. For fifteen years, he was Vice President of the International Institute of Space Law, after which he became an Honorary Director. Zhukov participated in international conferences, including sessions of the Legal Subcommittee of the United Nations Committee on Outer Space (1963, 1979) and the UN Committee on Outer Space (1979). He was part of the Working Group on direct broadcast satellites (1970) and the UN Conference on Outer Space (1968). In 1968, he was awarded a gold medal and diploma of the International Astronautical Federation and International Institute of Space Law for "Contribution to the Research Development of International Space Law."

[3] BIN CHENG, STUDIES IN INTERNATIONAL SPACE LAW (1997).

this issue by Russian lawyers.[4] Thus, among the many reasons for Russian legal experts and jurists to appreciate Professor Cheng's significance and work is the contribution of this outstanding British legal scholar to the formation and propagation of international space law doctrines that largely mirror Russian (and, formerly, Soviet) interpretations of international space law.

In his works, Professor Cheng considered a wide range of issues related to international space law, including a fairly detailed study of the problem of military uses of outer space. Today, not only are the problems related to the peaceful use of outer space still relevant, but, even more specifically, the issues of determining permissible limits for military activity and efforts to ban outer space from becoming a theater of war are waiting to be resolved.

The Russian doctrine related to international space law has traditionally emphasized the problem of demilitarization of outer space and space activities since the Soviet period of its history. Great contributions to the doctrinal foundations of space demilitarization have been made by Professor Zhukov,[5] Professor Yuri M. Kolosov,[6] and Judge Vladlen S. Vereshchetin.[7] Ongoing research in the field of international space law has further expanded the scope of this study.[8] In particular, there are works that propose the concept of international legal regulation of military space activities, although in the past, especially during the Soviet period, the topic was mainly focused on the prohibition of the militarization of outer space.[9] In this article, I will offer a modern understanding of the problem of demilitarization of outer space and space activities and the prospects for their international legal solution.

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[4] *See* GENNADY P. ZHUKOV & YURI M. KOLOSOV, INTERNATIONAL SPACE LAW 74 (Boris Belitzky trans., 2d ed. 2014).

[5] *See, e.g.*, GENNADY P. ZHUKOV, OUTER SPACE AND THE PEACE (1985).

[6] *See, e.g.*, YURI M. KOLOSOV, BOR'BA ZA MIRNYÎ KOSMOS [STRUGGLE FOR PEACEFUL SPACE: LEGAL ISSUES] (2d ed. 2014).

[7] *See generally* VLADLEN S.VERESHCHETIN, LEGAL ASPECTS OF PROHIBITING THE USE OF FORCE IN OUTER SPACE AND PREVENTING THE SPACE ARMS RACE//SPACE WEAPONS: THE SECURITY DILEMMA 172 (1986).

[8] *See e.g.*, Mikhail Nikolaevich Lysenko, Pravovye Problemy Perspektivy Zapreshcheniia͡ Oruzhiia͡ v Kosmose [Legal Problems and Prospects for Banning Weapons in Space] (2006) (dissertation, Moscow State University) (on file with author).

[9] See, e.g., Anatoly I. Antonov, Mezhdunarodno-pravovoe regulirovanie voenno-kosmicheskoĭ deia͡tel'nosti [International Legal Regulation of Military Space Activities], 4 MGIMO UNIV. BULL. 190, 196 (2012).

II. NUCLEAR DISARMAMENT AND DEMILITARIZATION OF OUTER SPACE

Initially, ideas about the need for international legal regulation of the activities of States in outer space were expressed in the period before the Second World War in the 1920s and 1930s. This was due to the development of aeronautics and its international legal regulation, which led to the birth of international air law. For advanced scholars of that era, it became clear that the next stage of development of the surrounding world would be space.[10] It is logical that the beginning of the space age, opened by the launch of *Sputnik I* on October 4, 1957, marked the beginning of the development of international documents, which were formulated drafts of normative regulation of conduct of States in space. The global nature of space exploration has determined the leading role of the United Nations (UN) in determining the legal strategy for this activity. The main ideas about the legal framework for international space regulation were set out in the form of resolutions of the United Nations General Assembly (UNGA). In the first decade of space exploration (1958-1967) the United Nations devoted itself to preparing principles for regulating space activities.

The beginning of the space age coincided with another equally significant era in the development of post-war international relations: the Cold War. The most characteristic feature of this time was the military-political rivalry between the two superpowers—the United States (US) and the Union of Soviet Socialist Republics (USSR or Soviet Union)—which affected all areas of their mutual relations. Space was no exception. Despite this rivalry, however, it was during the Cold War-era that the nuclear superpowers agreed to adopt international treaties that laid the international legal framework for limiting nuclear weapons and other weapons of mass destruction (WMD). The historical paradox was that the first Treaty norms related to outer space were enshrined in

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[10] Soviet scholars also contributed to pioneering work on space law. For example, in 1926, Valentin A. Zarzar, an expert in the field of international air law, suggested the development of "interplanetary transport law" in the future. *See, e.g.*, STEPHEN E. DOYLE, THE ORIGINS OF SPACE LAW AND THE INTERNATIONAL INSTITUTE OF SPACE LAW OF THE INTERNATIONAL ASTRONAUTICAL FEDERATION 2 (2002) (quoting Valentin A. Zarzar).

disarmament treaties—i.e. in sources of international law that were not directly related to the space sphere.

The Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space and Under Water (also known as the Moscow Treaty, or the Partial Test Ban Treaty) was adopted on August 5, 1963.[11] According to Article I(a) of the Moscow Treaty, each of its parties undertakes to prohibit, prevent and not produce any test explosions of nuclear weapons or any other nuclear explosions in any place under its jurisdiction or control in the atmosphere or beyond its borders, including outer space.[12]

Thus, even before the establishment of a special international legal regulation of space activities, mandatory Treaty norms were created within general international law prohibiting the testing of nuclear weapons in space.[13] This marked the beginning of the establishment of the regime of partial demilitarization of outer space and determined the vector of its further demilitarization.

The Soviet Union considered the nuclear demilitarization regime established in the Moscow Treaty to be insufficient, so in the same year, it proposed that the United States agree on a ban on placing objects with WMD on board in space.[14] This proposal was supported by the United States, and as a result, on October 17, 1963, the UNGA adopted Resolution 1884, "Question of General and Complete Disarmament."[15] This document welcomed the intention of the USSR and the United States to refrain from placing

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[11] Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space, and Under Water, Aug. 5, 1963, 14 U.S.T. 1313, 480 U.NT.S. 43. On September 10, 1996, during the 50th session of the UNGA, the Comprehensive Nuclear Test Ban Treaty was adopted by the United Nations. G.A. Res. 50/245 (Sept. 10, 1996). This Treaty extends the limited nuclear test ban regime to an unconditional framework. Taken together with the Moscow Treaty of 1963, these work to exclude nuclear explosions in Space.

[12] *Id.* at art. I(a).

[13] *See* G.A. Res. 1721 (XVI) (Dec. 20, 1961); G.A. Res. 1962 (XVIII) (Dec. 13, 1963). Indeed, all UNGA resolutions prior to the Outer Space Treaty that were devoted to space exploration, up to Declaration of Legal Principles Governing the Activities of States in the Exploration and Use of Outer Space Resolution of 1962, did not address issues of the militarization of outer space. *See* G.A. Res. 1962 (XVIII) (Dec. 13, 1963). In addition, it is no accident that the task of the standing committee on the Peaceful Uses of Outer Space, which was formed on the basis of UNGA Resolution 1721 (XVI), was to consider issues of international cooperation in the peaceful uses of outer space. *See* G.A. Res. 1721 (XVI) (Dec. 20, 1961).

[14] *See* ZHUKOV, *supra* note 5, at 30.

[15] G.A. Res. 1884 (XVIII) (Oct. 17, 1963).

objects with nuclear weapons or other types of WMD in orbit around the Earth.[16] Other States were called upon to refrain from placing any objects containing nuclear weapons or any other type of WMD in orbit around the Earth, installing such weapons on celestial bodies, or otherwise placing such weapons in outer space.[17] This resolution formulated a soft law rule limiting the proliferation of nuclear weapons and expanding the demilitarization of outer space.[18] At this stage of the development of the international legal regime of outer space, which was implemented in UNGA resolutions, the issues of the prohibition of nuclear weapons in outer space were not considered at all.

III. THE FORMATION OF INTERNATIONAL SPACE LAW AND THE DEMILITARIZATION OF OUTER SPACE.

The contours of international space law began to be defined immediately after the launch of *Sputnik I.*[19] In response to the initial Soviet success in space travel, the United States launched the *Explorer 1* artificial satellite on January 31, 1958.[20] First, within the framework of the UNGA, and then the UN Committee on the Peaceful Uses of Outer Space (COPUOS), the development of principles for space activities of States began. In particular, the USSR initiated the preparation of special international legal norms regulating the space activities of States, the implementation of broad and equal cooperation in the peaceful exploration of space, and the creation of the necessary bodies for these purposes.[21] Passed in December 1963, Resolution 1962 entitled "Declaration of Legal Principles Governing the Activities of States in the Exploration and Use of Outer Space," which was considered as the basis for the future Outer Space Treaty, practically did not address the issues of

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[16] *Id.* ¶ 1.

[17] *Id.* ¶ 2.

[18] *Id.* "Soft law" is catchall of non-binding, yet persuasive, sources of international law including United Nations General Assembly resolutions. *See Hard Law / Soft Law*, EUROPEAN CTR. CONST. & HUM. RTS., https://www.ecchr.eu/en/glossary/hard-law-soft-law/ (last visited Oct. 13, 2020).

[19] *See* Vladimír Kopal, *Evolution of the Doctrine of Space Law*, *in* SPACE LAW: DEVELOPMENT AND SCOPE 17, 19 (Nandasiri Jasentuliyana ed., 1992).

[20] *Explorer 1 Overview,* NASA, https://www.nasa.gov/mission_pages/explorer/explorer-overview.html (last visited Oct. 13, 2020).

[21] *See* G.A. Res. 1472 (XIV) (Dec. 12, 1959).

military use of outer space.[22] The only thing that was included in its preamble was the condemnation of propaganda aimed at or capable of creating or increasing a threat to the peace, violation of the peace or acts of aggression.[23] This rule was contained in Resolution 110 of November 3, 1947, and its mention in the preamble of the Declaration meant extending the condemnation of pro-militarist propaganda to outer space.[24]

It can be said that the lack of provisions on the military aspect of space activities was not an accidental gap, but rather the result of disagreements between the two space powers over the strategy for the military use of space. Therefore, the UNGA limited itself to fixing the principle establishing the extension of the norms of international law to outer space, including the UN Charter, which established the political and legal framework for ensuring sovereignty and mechanisms for collective measures to counter aggression.[25]

The first treaty to address the international legal implications of outer space was the 1967 Outer Space Treaty.[26] This document has several features in terms of establishing legal means to prevent the militarization of outer space. First of all, it establishes the obligation of its participants to carry out activities in the exploration and use of outer space, including the Moon and other celestial bodies, in accordance with international law, including the UN Charter, in the interests of maintaining international peace and security and promoting international cooperation and understanding.[27] Thus, space activities should be based on international law and the UN Charter and contribute to the achievement of the UN goals in maintaining international peace and security and promoting international cooperation.[28] It should be recognized that the Outer Space Treaty obligations are formulated in very general terms, but

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[22] *See* G.A. Res. 1962 (XVIII) (Dec. 13, 1963).

[23] *Id.* at Preamble.

[24] G.A. Res. 110 (II) (Nov. 3, 1947).

[25] U.N. Charter art. 2, ¶ 4 (requiring that all members refrain from the threat or use of force in their international relations against the territorial integrity or political independence of any State); U.N. Charter art. 51 ("[n]othing in the present Charter shall impair the inherent right of individual or collective self-defense").

[26] Outer Space Treaty, *supra* note 1.

[27] *Id.* at art. III.

[28] U.N. Charter art. 1.

nevertheless they regulate the implementation of space activities in accordance with the most important goals of the United Nations.

Article IV of the Outer Space Treaty establishes the partial demilitarization of outer space forbidding: the placement in orbit around the Earth of any objects with nuclear weapons or any other kinds of WMD; the installment of such weapons on celestial bodies; and the stationing of such weapons in outer space in any other manner.[29] This provision gives legal effect to the preamble to the Outer Space Treaty, which refers to the above-mentioned Resolution 1884.[30] At the same time, the preamble refers to the above-mentioned Resolution 110, but the ban on militaristic propaganda itself is not reflected in the articles of the Outer Space Treaty.[31]

Article IV of the Outer Space Treaty also establishes a complete demilitarization regime for the Moon and other celestial bodies.[32] It provides that the Moon and other celestial bodies are to be used by all States parties to this treaty "exclusively for peaceful purposes."[33] Article IV additionally prohibits "the establishment of military bases, installations and fortifications, the testing of any type of weapons and the conduct of military maneuvers on celestial bodies."[34] However, "the use of military personnel for scientific research or any other peaceful purposes is not prohibited."[35] It is also not prohibited to use "any equipment or facilities necessary for the peaceful exploration of the moon and other celestial bodies."[36]

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[29] Outer Space Treaty, *supra* note 1, art. IV. The inclusion of Article IV in the Outer Space Treaty allowed Kai-Uwe Schrogl and Julia Neumann to conclude that this Treaty is clearly an arms control treaty, which was the result of the authors' awareness of the need to ban nuclear weapons and other types of WMD in outer space. Kai-Uwe Schrogl, Julia Neumann, *Article IV*, *in* COLOGNE COMMENTARY ON SPACE LAW 72 (Hobe et al. eds., 2009). Does this mean that the Outer Space Treaty is hybrid in nature, that is it regulates both disarmament and space law issues simultaneously, or is the inclusion of provisions on the demilitarization of outer space, the Moon and other celestial bodies intended to recognize the special nature and significance of space disarmament? This question remains unanswered.

[30] G.A. Res. 1884 (XVIII) (Oct. 17, 1963) (calling upon States to "refrain from placing in orbit around the [E]arth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction, [or] from installing such weapons on celestial bodies . . .").

[31] *See* G.A. Res. 110 (II) (Nov. 3, 1947) (condemning propaganda designed or likely to provoke or encourage any threat to the peace).

[32] Outer Space Treaty, *supra* note 1, art. IV.

[33] *Id.*

[34] *Id.*

[35] *Id.*

[36] *Id.*

Professor Cheng supported the provision on establishing a complete demilitarization regime for the Moon and other celestial bodies in Article IV of the Outer Space Treaty.[37] He argued that the provision in question should be interpreted in the same way as Article 1 of the 1959 Antarctic Treaty,[38] namely, no military activity for military purposes should be conducted on the Moon and other celestial bodies.[39]

The international legal regime for the demilitarization of Outer Space is not addressed in the subsequent three agreements on space-related activities: the 1968 Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space,[40] the 1972 Convention on International Liability for Damage Caused by Space Objects[41] and the 1974 Convention on Registration of Objects Launched into Outer Space.[42] However, these treaties all promote international cooperation in the peaceful exploration and use of outer space.

In addition, the legal regime for the demilitarization of outer space has been expanded to include provisions of other multilateral and bilateral international agreements related to disarmament. For example, the 1977 Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques applies and "refers to any technique for changing – through the deliberate manipulation of natural processes – the dynamics,

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[37] *See generally* Bin Cheng, *Le Traite de 1967 sur l'espace*, 36 J. DU DROIT INT'L 608 (1968).

[38] The Antarctic Treaty art. 1, Dec. 1, 1959, 12. U.S.T. 794, 42 U.N.T.S. 71.

[39] *See* Cheng, *supra* note 37.

[40] Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space, Apr. 22, 1968, 19 U.S.T. 7570, 672 U.N.T.S. 119.

[41] Convention on the International Liability for Damage Caused by Space Objects, Mar. 29, 1972, 24 U.S.T. 2389, 961 U.N.T.S. 187. The possibility of applying the concept of absolute liability adopted by this Convention in order to compensate for damage to targets located on Earth and in airspace as a result of military attacks from weapons deployed in space has been discussed. We should agree with Professor Sa'id Mosteshar that the concept of absolute responsibility applies to belligerents only in cases of damage caused in violation of international humanitarian law. *See* Sa'id Mosteshar, *Space Law and Weapons in Space*, OXFORD RSCH. ENCYCLOPEDIA - PLANETARY SCI. (May 23, 2019), https://oxfordre.com/planetaryscience/view/10.1093/acrefore/9780190647926.001.0001/acrefore-9780190647926-e-74.

[42] Registration of Objects Launched into Outer Space, Jan. 14, 1975, 28 U.S.T. 695, 1023 U.N.T.S. 15.

composition or structure . . . of outer space."[43] The Convention prohibits the use of environmental modification techniques for military purposes.[44] Additionally, a number of provisions on the restriction of military activities in outer space were contained in the Soviet-American agreements in the field of disarmament. For example, the USSR-US Strategic Arms Limitation Treaty of 1979 (SALT II) established a ban on partially orbiting missiles, that is, missiles that do not follow an orbital trajectory, but are placed in and removed from orbit in order to strike an object on Earth.[45] While SALT II has not entered into force, the treaty was actually executed by the parties, evincing an international recognition of disarmament interests. For its part, the 1972 Treaty on the Limitation of Anti-Ballistic Missile Systems obligated the parties not to create, deploy or test anti-ballistic missile systems as space-based components.[46]

The provisions on the demilitarization of the Moon and other celestial bodies established in the Outer Space Treaty were substantially developed in the Agreement Concerning the Activities of States on the Moon and Other Celestial Bodies of December 5, 1979[47] (Moon Agreement). In particular, the Moon Agreement significantly expands the requirements for demilitarization set out in Article IV(2) of the Outer Space Treaty.[48] First of all, the Preamble notes the desire of the parties to the Moon Agreement to prevent the Moon "from becoming an area of international conflict."[49] Article 2 of the Moon Agreement emphasizes that activities on the Moon, including its exploration and use, are

> [C]arried out in accordance with international law, in particular the UN Charter, and taking into account the Declaration on Principles of International Law concerning Friendly

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[43] Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques art. II, Dec. 10, 1976, 1108 U.N.T.S. 151, 31 U.S.T. 333.

[44] *Id.* at art. I.

[45] Treaty Between the United States of America and the Union of Soviet Socialist Republics on the Limitation of Strategic Offensive Arms (SALT II), U.S.-U.S.S.R., June 18, 1972, S. TREATY DOC. NO. 96-1 (1979), 18 I.L.M. 1138.

[46] Treaty Between the United States of America and the Union of Soviet Socialist Republics on the Limitation of Anti-Ballistic Missile Systems art. V, U.S.-U.S.S.R., May 26, 1972, 23 U.S.T. 3435, 944 U.N.T.S. 13 (entered into force Oct. 3, 1972).

[47] Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, Dec. 18, 1979, 1363 U.N.T.S. 3 [hereinafter Moon Agreement].

[48] Outer Space Treaty, *supra* note 1, art. IV.

[49] Moon Agreement, *supra* note 47, Preamble.

> Relations and Cooperation among States in accordance with the UN Charter, adopted by the UNGA on 24 October 1970, in the interests of maintaining international peace and security and promoting international cooperation and understanding, and with due regard to the relevant interests of all other States parties.[50]

Thus, the relevant provisions of the UN Charter prohibiting the use of force or the threat of force[51] on the Moon and other celestial bodies should be understood in accordance with the provisions of the 1970 Declaration on Principles of International Law.[52]

Article 3 of the Moon Agreement repeats the provisions of the Outer Space Treaty on the neutralization and demilitarization of the territory of this natural satellite of the Earth.[53] The provision on the use of the Moon exclusively for peaceful purposes is confirmed, which underlines the importance of this element of its international legal regime.[54] The Article specifically mirrors the provisions of Article IV of the Outer Space Treaty prohibiting the creation of military bases, structures and fortifications on celestial bodies, testing of any type of weapons and conducting military maneuvers.[55] It confirms the permissibility of using military personnel for scientific research or any other peaceful purposes, as well as the use of any equipment or facilities necessary for peaceful research, not only as provided for in Article IV of the Outer Space Treaty, but also for the use of the Moon.[56] This last clarification seems logical, especially in light of the planned lunar missions in the coming decade.

The nuclear-free regime of outer space has been further developed. States parties have undertaken not to place objects containing nuclear weapons or any other WMD in orbit around the Moon

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[50] Moon Agreement, *supra* note 47, art. 2 (citations omitted). The Declaration on Principles of International Law concerning Friendly Relations and Cooperation among States in accordance with the UN Charter, adopted by the UNGA on 24 October 1970 can be found at G.A. Res. 2625 (XXV) (Oct. 24, 1970).

[51] U.N. Charter art. 2.

[52] G.A. Res. 2625 (XXV) (Oct. 24, 1970).

[53] Moon Agreement, *supra* note 47, art. 3.

[54] *Id.* at art. 3(1).

[55] *Id.* at art. 3(3)-(4); Outer Space Treaty, *supra* note 1, art. IV.

[56] Moon Agreement, *supra* note 47, art. 3(4).

or on any other flight path to or around the Moon, nor to install or use such weapons on the surface of the Moon or in its interior.

At the same time, the demilitarization regime of the Moon is supplemented by important provisions prohibiting the threat or use of force or any other hostile action or threat to commit hostile actions on it. The concept of prohibiting the threat or use of force in outer space is consistent with the position set out in the Russian-Chinese joint draft Treaty on the Prevention of the Placement of Weapons in Outer Space, the Use of Force or the Threat of Force Against Space Objects[57] (PPWT), but does not coincide with it word for word.[58] The Moon Agreement also prohibits the use of the Moon to perform any hostile actions or use any threat of hostile action against the Earth, the Moon, spacecraft, spacecraft personnel or artificial Space objects.[59] These restrictions and prohibitions allow States to avoid possible conflicts when conducting research and developing natural resources on the Moon, which may become a reality in the very near future.

However, the authority of the established international legal regime for the demilitarization of the Moon, its orbit or other flight path to or around the Moon is weakened not by the quality of the legal language used, but by the small number of States participating in the Moon Agreement (eighteen States have ratified or acceded, and four States have signed, among the latter include two spacefaring powers – France and India).[60] It is the author's view

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[57] Draft Treaty on Prevention of the Placement of Weapons in Outer Space and of the Threat or Use of Force Against Outer Space Objects, Feb. 12, 2008. The Russian Federation and China jointly submitted this document to the Conference on Disarmament (CD) in Geneva. In June 2014, Russia and China submitted an updated draft of the PPWT to the CD, taking into account the proposals made by interested States since its introduction. Draft Treaty on Prevention of the Placement of Weapons in Outer Space and of the Threat or Use of Force Against Outer Space Objects, June 10, 2014. The draft can be found at: reachingcriticalwill.org/images/documents/Disarmament-fora/cd/2014/documents/PPWT2014.pdf [hereinafter PPWT 2014].

[58] The "use of force" or "threat of force" in the updated draft of the 2014 PPWT refers to any deliberate action aimed at causing damage to a space object under the jurisdiction and/or control of other States, or an explicit intention to commit such an action in writing, orally or in any other form. Actions in accordance with separate agreements signed with such States that provide for actions to terminate the unguided flight of a Space object under their jurisdiction and/or control at the request of such States cannot be regarded as the use of force or the threat of force. PPWT 2014, *supra* note 57, art. I(d).

[59] Moon Agreement, *supra* note 47, art. 3(2).

[60] *See* STATUS OF INTERNATIONAL AGREEMENTS RELATING TO ACTIVITIES IN OUTER SPACE AS AT 1 JANUARY 2020, U.N. OFF. OUTER SPACE AFF. (2020),

that the reasons for such a narrow circle of participants in the Moon Agreement are due to the lack of balance in the positions of the States involved in its preparation on the key issue of the legal regime of the Moon's resources. The legal regime of the Moon and other celestial bodies was established at the insistence of developing countries. According to Article 11 of the Moon Agreement, the Moon and other celestial bodies, as well as their natural resources, are the "common heritage of [hu]mankind," and the exploitation of such resources should be regulated by an international regime (including appropriate procedures), which States will establish when it is obvious that such exploitation will become possible in the near future.[61] Moreover, the main goals of the future international regime listed in Article 11 are not supported by the industrialized countries.[62]

It follows that the reasons for the non-participation of the leading spacefaring powers in the Moon Agreement lie in a different plane than the demilitarization of the Moon and other celestial bodies. However, we cannot ignore the fact that, unlike the other four space treaties, the Moon Agreement has not yet become universal, which is becoming the subject of expert discussions about finding acceptable solutions.[63]

IV. PROSPECTS FOR STRENGTHENING THE DEMILITARIZATION OF OUTER SPACE REGIME

The legal basis of the international legal regime for the demilitarization of outer space has undergone certain metamorphoses in the last decades of the twentieth century and in the twenty-first century. In the period of the 1960s-1970s, the main means of its

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https://www.unoosa.org/documents/pdf/spacelaw/treatystatus/TreatiesStatus-2020E.pdf.

[61] Moon Agreement, *supra* note 47, art. 11.

[62] *Id.* at art. 11(7). *See also* P.A. Berkman et al., *Russia-United States Common Challenges and Perspectives,* 1 MOSCOW J. INT'L L. 16, 24 (2018) (analyzing the international legal regime of resources of the Moon and celestial bodies established by the Moon Agreement from the point of view of Russian and American legal doctrines).

[63] A number of experts consider it necessary to revise this agreement in accordance with a 1994 revision of the UN Convention on the Law of the Sea of 1982. *See* Gennady P. Zhukov, *Mezhdunarodnoe Pravo i Vyzovy XXI Veka* [*International Space Law and Challenges of the 21st Century*], *in* MEZHDUNARODNOE PRAVO: UCHEBNIK DLIĀ ASPIRANTOV [INTERNATIONAL LAW: TEXTBOOK FOR GRADUATE STUDENTS] 430 (A.K. Abashidze ed., 2018).

formation was the treaty regulation of either acts of international disarmament law or international space treaties. At the same time, UNGA resolutions on space activities were assigned the role of secondary, auxiliary regulators that form the contours of future treaty norms. Since the beginning of the 1980s, the situation has changed, and the international treaty has become a tool for developing bilateral and multilateral cooperation between States in the field of space exploration. At the universal level, there is an expansion of the significance of the UNGA resolutions, which formed the main tracks for preventing the militarization of outer space.

In parallel, there are examples of the development of "soft regulation" documents, which are aimed not at achieving the goal of carrying out space activities "exclusively for peaceful purposes," but at improving the security conditions in outer space. The European Union (EU) introduced a draft Code of Conduct in 2008.[64] It notes that the draft Code contains fundamental rules that should be observed by all States conducting space activities to, among other things, prevent an arms race in outer space.[65] In addition, it is aimed at reducing the risk of collisions and the appearance of space debris, as well as strengthening mutual understanding between States engaged in space activity.[66]

At the same time, there are initiatives to further strengthening the Treaty framework for the demilitarization of outer space. In particular, the Russian Federation prefers the formation of a Treaty-based legal mechanism for ensuring the safety of space activities and preventing an arms race in Outer Space. We are talking about promoting the initiative of the above-mentioned Russian-Chinese PPWT, which provides for a ban on the placement of weapons of any kind in space and on any use of force against space objects.[67]

From the point of view of the logic of preventing an arms race in outer space, it looks like a consistent continuation of the regime of partial demilitarization of outer space already enshrined in

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[64] EUROPEAN UNION - EUROPEAN EXTERNAL ACTION SERVICE, DRAFT INTERNATIONAL CODE OF CONDUCT FOR OUTER SPACE ACTIVITIES (Mar. 31, 2014), http://www.eeas.europa.eu/archives/docs/non-proliferation-and-disarmament/pdf/space_code_conduct_draft_vers_31-march-2014_en.pdf.

[65] *Id.* at Preamble.

[66] *Id.* at arts. 4-5. *See* Zhukov, *supra* note 63, at 418.

[67] *Id.* at 407-18.

international space law.[68] The extension of the ban on the deployment of any weapons in outer space, not just nuclear weapons or WMD, meets modern realities. A special international treaty prohibiting the use of force or the threat of force against space objects in addition to the mandatory requirements of Article 2 of the UN Charter[69] also deserves support. The solidification and development of the non-use of force or the threat of force in relation to objects located in outer space is quite justified, since this would be the first concrete step towards a legal settlement of the threat of using anti-satellite systems that concerns most spacefaring States. Such a step, which is legitimate from the point of view of Article 1(4) of the UN Charter, would significantly improve the atmosphere of distrust that currently reigns in relations between the leading space powers.[70]

Without detracting from the importance of "soft regulation" tools for ensuring the security of States' space activities, I present a number of considerations on the need and urgency of a contractual method for strengthening the demilitarization and weaponization of outer space. First of all, support should be given to the claim that the current Outer Space Treaty of 1967 in terms of the ban on the deployment of nuclear and other weapons of mass destruction, it has been a very effective obstacle to an arms race in outer space. Indeed, no violations of the regime established by Article IV of the Outer Space Treaty have been observed during its operation. Moreover, despite some doubts about its potential for future regulation of space activities,[71] a significant number of international lawyers give it a high assessment.[72]

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[68] Outer Space Treaty, *supra* note 1, art. IV; Moon Agreement, *supra* note 46, arts. 2-3.

[69] *See* U.N. Charter art. 2.

[70] *Id.* at art. 1, ¶ 4 ("To be a center for harmonizing the actions of nations in the attainment of these common ends.").

[71] Ram S. Jakhu, *Dogovoru po Kosmosu 1967 Goda Ispolniâetsiâ 40 Let* [*The 1967 Outer Space Treaty at its 40th Anniversary*], *in* SOVREMENNYE PROBLEMY MEZHDUNARODNOGO KOSMICHESKOGO PRAVA [MODERN PROBLEMS IN INTERNATIONAL SPACE LAW] 202-03 (Gennady Zhukov & Anatoly Kapustin eds., 2008).

[72] Some scholars call it the Constitution of Outer Space. *See* M.T. Andem, *Договор по космосу 1967 года как Великая хартия современного космического права: краткое размышление* [*The 1967 Outer Space Treaty as the Magna Carta of Contemporary Space Law: A Brief Reflection*], *in* SOVREMENNYE PROBLEMY MEZHDUNARODNOGO KOSMICHESKOGO PRAVA [MODERN PROBLEMS IN INTERNATIONAL SPACE LAW] 177 (Gennady Zhukov & Anatoly Kapustin eds., 2008); Jakhu, *supra* note 71.

Professor Zhukov notes that "the Outer Space Treaty is the most successful product of law-making within the framework of the United Nations in the twentieth century,"[73] which gives grounds for confidence that in the foreseeable future it will play a fundamental role in maintaining international law and order in outer space. The Outer Space Treaty with regard to the development of the provisions of Article IV, as shown by the Moon Agreement, does not create obstacles to the expansion and strengthening of the demilitarization regime on the Moon and celestial bodies and can serve as the basis for the further development of this regime in orbit around the Earth or anywhere in outer space, or on any celestial body other than the Earth.

We should agree that the importance of the treaty settlement of space weapons increases due to the lack of international legal guarantees not to launch weapons into space. In addition, the lack of such guarantees contributes to increasing uncertainty about the strategic capabilities and intentions of the space powers, which are increasing in number. The deployment of non-WMD weapons in outer space would have a global area of operation, high readiness for use, and the ability to covertly affect and disable space and ground objects. It would become a weapon of real use, generate suspicion and tension in relations between States, and disrupt the climate of mutual trust and cooperation in peaceful space exploration.[74]

Finding a solution to the potential problem of deployed weapons in outer space, and prohibiting the use or threat of force against space objects, are not the only way to strengthen the long-term sustainability of space activities. This question deserves a separate study. It is important to emphasize that such efforts are consistent with both the basic principles of international law and the UN Charter, as well as the goals of peaceful and safe space activities.

Of course, the choice of one or another method of legal settlement of the problem is not a panacea for the risks and threats that exist in inter-State relations in outer space. However, a violation of an international treaty entails international responsibility not only on the basis of international space law, but also by virtue of general

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[73] Zhukov, *supra* note 63, at 418.

[74] *See id.* at 406-07.

international law. This fact can play a role in making foreign policy decisions by individual States.

V. CONCLUSION

An analysis of the formation and development of international legal regulation of the establishment of the demilitarization regime in outer space shows that this process began with the inclusion of relevant provisions in treaties on the control of nuclear weapons and other types of WMD. Initially, the UNGA resolutions devoted to the development of goals and principles for regulating the behavior of States in outer space were silent on "disarmament" issues. However, the Outer Space Treaty established a mandatory regime for the partial demilitarization of outer space and the complete demilitarization of the Moon and other celestial bodies. The Moon agreement has reinforced this positive trend, although it has not gained universal significance

At present, efforts to prevent an arms race in outer space are a key focus of the world community's attention. In this regard, the UNGA attaches great importance to transparency and confidence-building measures in outer space, as well as to further practical measures to prevent an arms race in outer space, and, indeed, to prevent the first deployment of space weapons. UNGA resolution 74/34, "Further practical measures for the prevention of an arms race in Outer Space," of 12 December 2019, recognizes that although existing international treaties relating to the outer space legal regime play a positive role in regulating space activities, they are not able to completely prevent the deployment of weapons in outer space.[75] Thus, the existing treaty regime is incapable of preventing an arms race in outer space, and there is a need to strengthen and consolidate this regime. As a result of the study, it was concluded that the most optimal and reliable means of achieving the goal of keeping outer space peaceful and demilitarized is the development of an appropriate international Treaty better regulating outer space weapons, a draft of which was proposed jointly by Russia and China.[76]

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[75] G.A. Res. 74/34 (Dec. 12, 2019).

[76] *See* PPWT 2014, *supra* note 57.